

Schedule to Section 63 of the Bharatiya Sakshya Adhiniyam, 2023

Part B — Technical Expert Certification (Section 79A, IT Act authorised expert)

Expert Name & Designation	Dr. Authorized Forensic Examiner — Digital Forensic Examiner (Sec 79A IT Act)
Expert ID / Licence	IT-79A-IND-2026-XXXX
Verification Workstation Serial	BHARATSHIELD-LOCAL-001
Verification MAC	00:00:00:00:00:00
Independent SHA-256 Confirmation	b43ea392960d67ed418657ff6b61399cfd641e2996a6 (verified match)
Date & Time of Examination (IST)	03-06-2026 15:55:04 IST

Forensic Analysis Summary

The electronic record identified above was extracted using write-protected forensic procedures. Structural integrity of the bitstream was preserved. No transcoding or lossy re-encoding was applied during acquisition.

- Spatial CNN manipulation probability: **0.9041**
- 3D face mesh landmark variance: **0.099848**

- Lip-sync alignment error: **149.82 ms**
- TTS synthetic voice probability: **0.9985**
- Spectrogram pitch mismatch ratio: **0.4994**
- Anti-spoofing NN confidence (bona fide): **0.0012**

Resolved Subject: Subject Under Investigation — source: user_override

Grounds for Deepfake Classification

Summary: Media classified as likely synthetically manipulated based on 7 forensic indicator(s), including 4 high-severity finding(s).

- **Audio-Video Sync (critical):** Audio–video lip movement is misaligned, consistent with synthetic dubbing or post-hoc voice replacement rather than naturally captured speech. (*lip_sync_alignment_error_ms: 149.8 ms*)
- **Facial Manipulation (high):** Spatial facial inconsistencies detected across frames suggest GAN or deepfake-based facial manipulation. (*spatial_cnn_manipulation_probability: 90.41%*)
- **3D Face Geometry (medium):** 3D face mesh landmarks are unstable across the temporal sequence, indicating inconsistent facial geometry typical of synthetic faces.

(face_mesh_landmark_variance: 0.0998)

- **Synthetic Voice (high):** The voice track exhibits characteristics consistent with text-to-speech or neural voice cloning rather than a live recording.
(tts_synthetic_probability: 99.85%)
- **Voice Authenticity (medium):** Pitch contour and spectrogram features do not match natural speech patterns for the apparent speaker, suggesting audio synthesis or splicing.
(spectrogram_pitch_mismatch_ratio: 49.94%)
- **Anti-Spoofing (high):** Anti-spoofing neural network assigns low confidence to bona fide speech, indicating the audio is likely non-genuine or heavily processed.
(anti_spoofing_nn_confidence: 0.12%)
- **Analyst Observation (medium):** Auto-generated from BharatShield detection.
CNN=0.9041, FFT=0.9985, Verdict=Fake

I certify that the above analysis was conducted on the hash-verified original and that the opinions expressed are based on validated multi-modal deepfake detection models.

Expert Signature: _____ Date: 03-06-2026